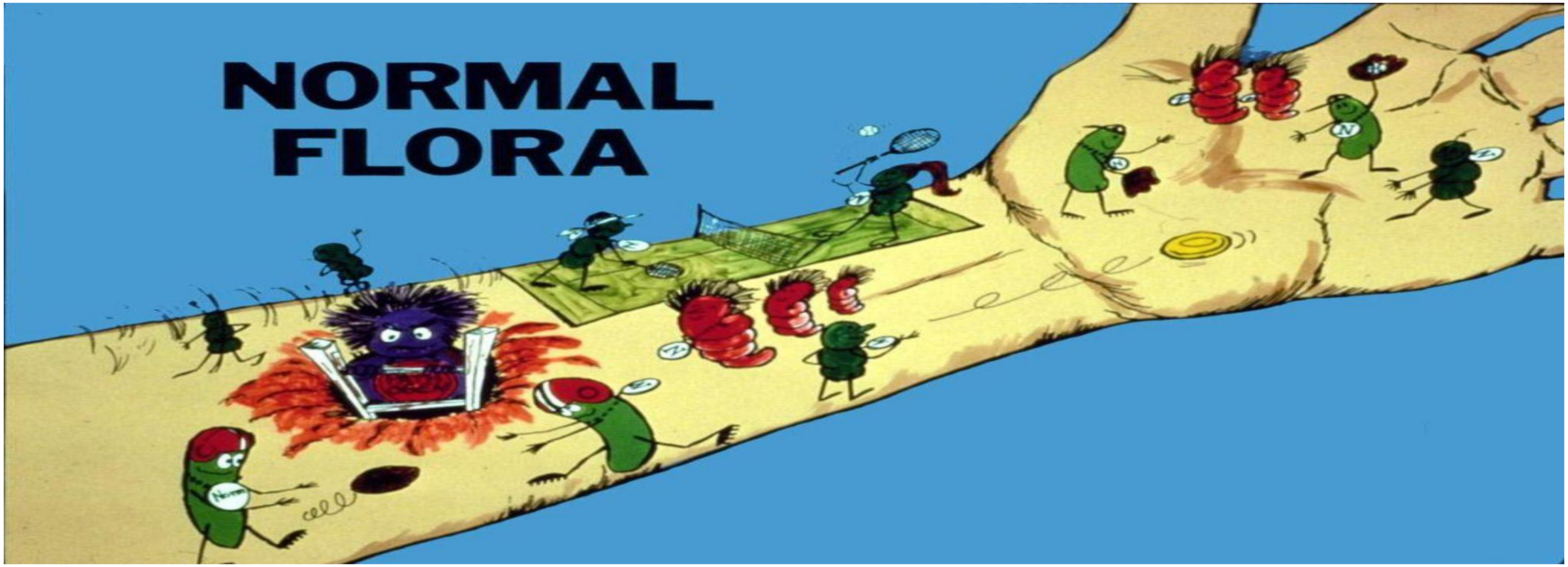
Normal Microbial Flora
Normal Microbiota
Microbiome



Assist.prof Dr. Vian B. Barzinjy
MSc, PhD.Microbiology
Hawler Medical University
College of Medicine

Objectives

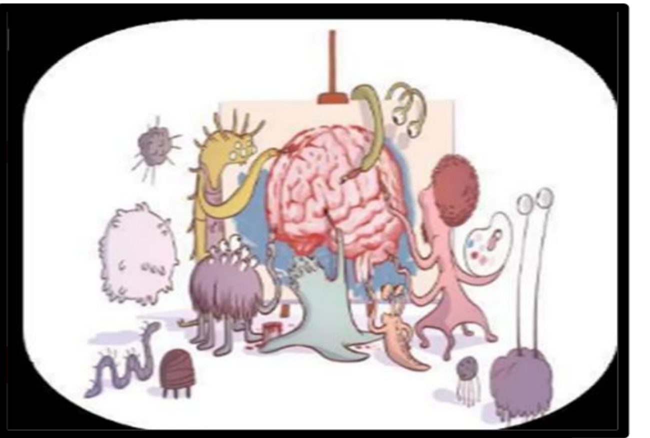
1- Definition.

2-Human body sites and normal microbial flora.

3-Advantageous and disadvantageous of normal flora

4-Types of normal flora

What is Normal Flora



- These are mixture of microorganisms(bacteria & fungi) regularly found at any anatomical site on /within the body of a healthy person.
- The viruses and parasites are usually not considered members of the normal flora although they can be present in asymptomatic individuals.
- Some of these microorganisms are found in association with humans / animals only.
- Others are found in the environment as well.
- More bacterial than human cells in the body founds.

Where on a healthy human is the microbiome located ?

Every human body surface which is exposed to the environment and every body part with an opening to the environment has a microbiome.

Sites that harbor a normal flora :

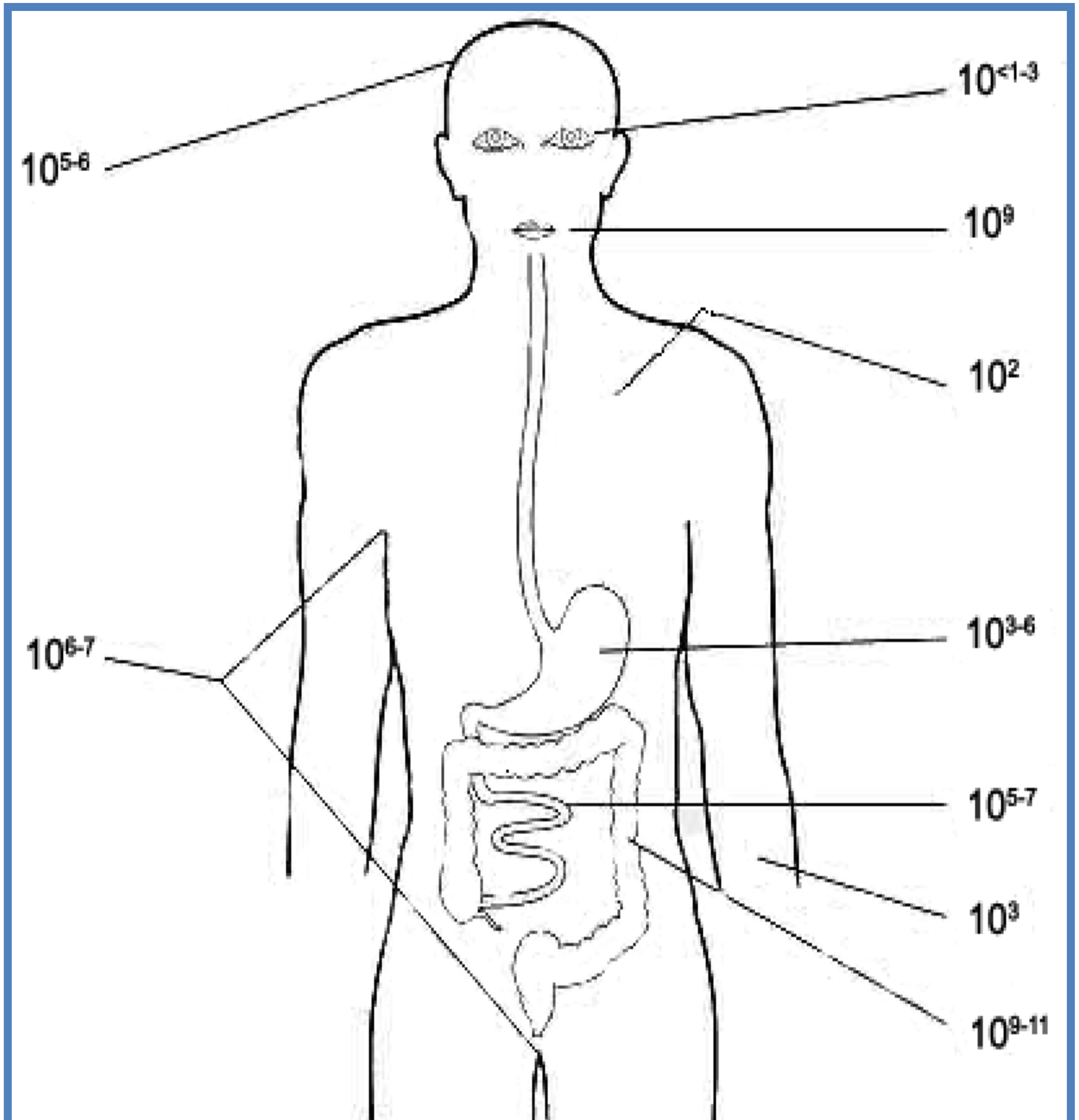
- Skin & mucous membranes.
- External ear canal.
- Upper respiratory tract
- Gastroi ntestestinal tract.
- Outer opening of urethra.
- External genitalia.
- External eye(Lids,Conjuctiva)
- Mouth, (human oral cavity).

WHEN WE GET COLONIZED WITH NORMAL FLORA

- A human first becomes colonized by a normal flora at the moment of birth and passage through the birth canal. In utero, the fetus is sterile,

  when birth process begins, so does colonization of the body surfaces. Handling and feeding of the infant after birth leads to establishment of a stable normal flora on the skin, oral cavity and intestinal tract in about 48 hours.

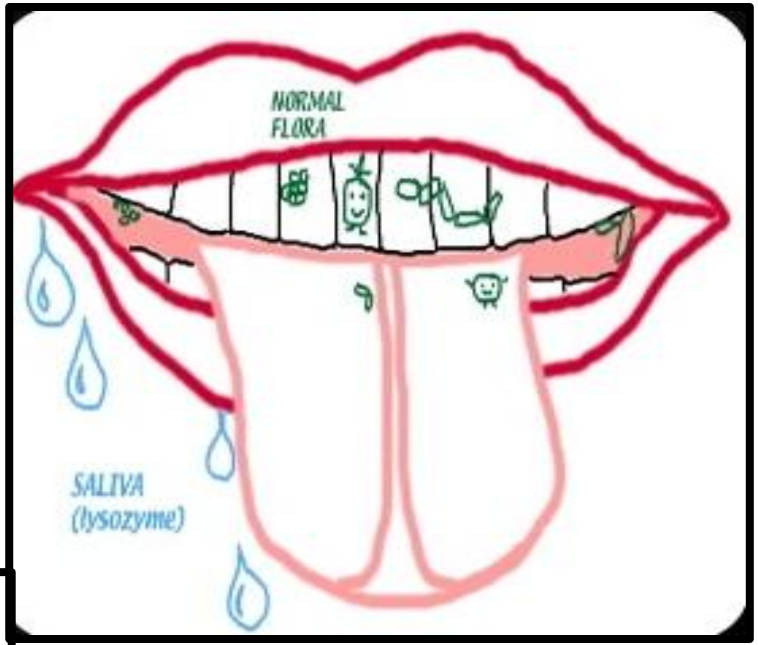
Estimation of the Normal flora

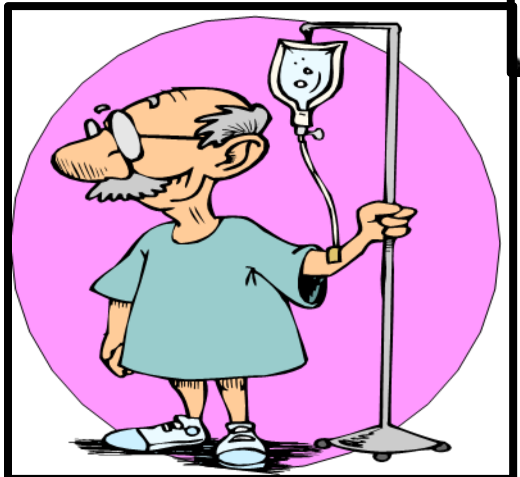


- **Human body**
- **10^{13} cells**
- **10^{14} bacteria**
- **Some bacteria occupy more than one niche**
- **Some bacteria occupy only one niche: tissue tropism**

Factors Influencing Normal Flora:

1. Age
2. Diet
3. Health condition(immunity)
3. Local environment
   (PH,temp.,H2o & nutrient
   level)
4. The use of antibiotics





A NEW BORN CHILD'S FLORA IS DEPENDENT ON MOTHER

- The composition of a child's bacterial flora is dependent on the mother's micro flora, since she is the primary source for the child's bacteria at the outset

Protective function (barrier effect):

- Compete & adhere to the attachment sites in the brush border of intestinal epithelial.
- Compete for available nutrients.
- Produce antimicrobial (bacteriocins).

All of this will prevent attachment and subsequent entry of pathogenic bacteria into the epithelial cells.

- **Sterile tissues:**

- In a healthy human, the internal tissues such as:
- ❑ Blood
- ❑ Brain
- ❑ Muscle
- ❑ Cerbrospinal fluid(cs.f)

Are normally free of microorganisms


We Are Really More Bug than Man.......
Humans as micro biomes:-
- 10-100 trillion microbes in human intestine.
- 3 million genes (100X).
- 2 kg weight.
- 300-1000 species of bacteria.
- control almost all body functions.
100 % Human?
90% microbes
10% human cells

Tissue specificity of normal flora

Most members of normal bacterial flora prefer to colonize certain tissues and not others. This tissue specificity is, Usually due to properties of both the host and the bacterium
Usually ,specific bacteria colonize specific tissues by one or another of these mechanisms.

1-Tissue tropism. 2- Specific adherence. 3-biofilm formation.

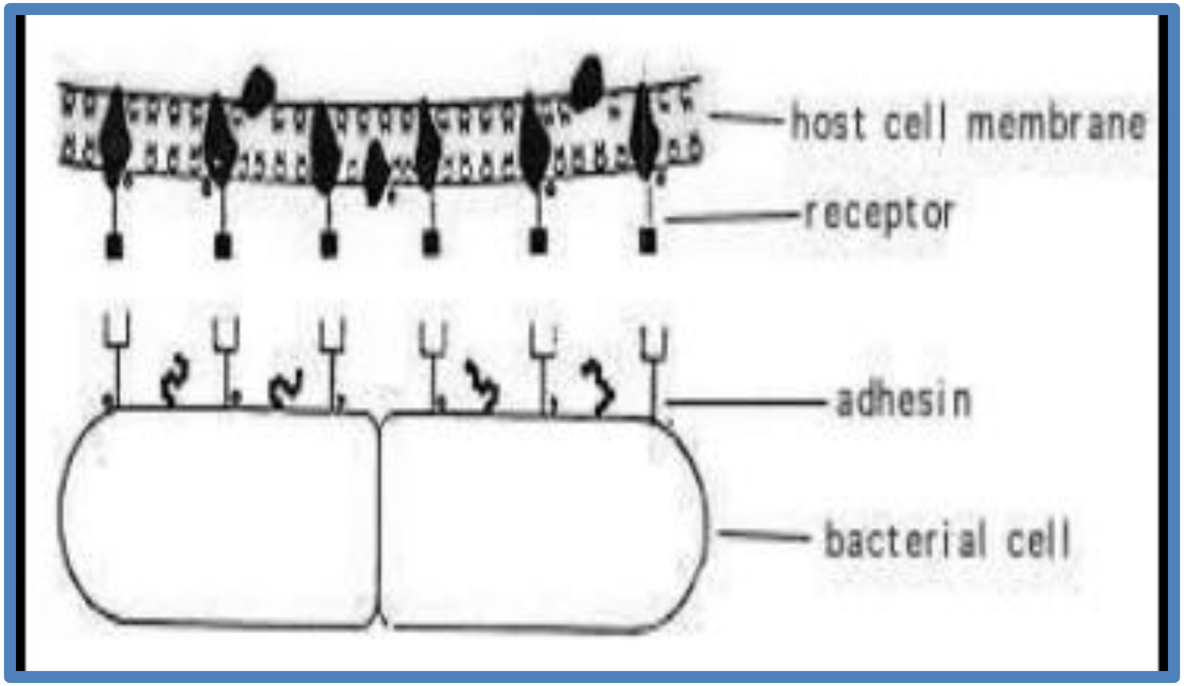
1-Tissue tropism :

_is the bacterial preference for certain tissues for growth .One explanatation for tissue tropism is that the host provides essential nutrients and growth factors for the bacterium, in addition to suitable oxygen.PH,and temperature for growth.

For eg.:*Lactobacillus acidophilus*. informally knowns as " doderleins bacillus" colonizes the vagina because glycogen is produced which provides the bacteria with a source of sugar that they ferment to lactic acid.

2- Specific adherence:

-Most bacteria can colonize in specific tissue because they can adhere to that tissue in a specific manner that involves chemical interactions between bacterial surface components and host cell molecular receptors. The bacterial components are molecular parts of their capsules, fimbriae, or cell walls . The receptors on human cells are usually glycoprotein molecules located on the host cell.



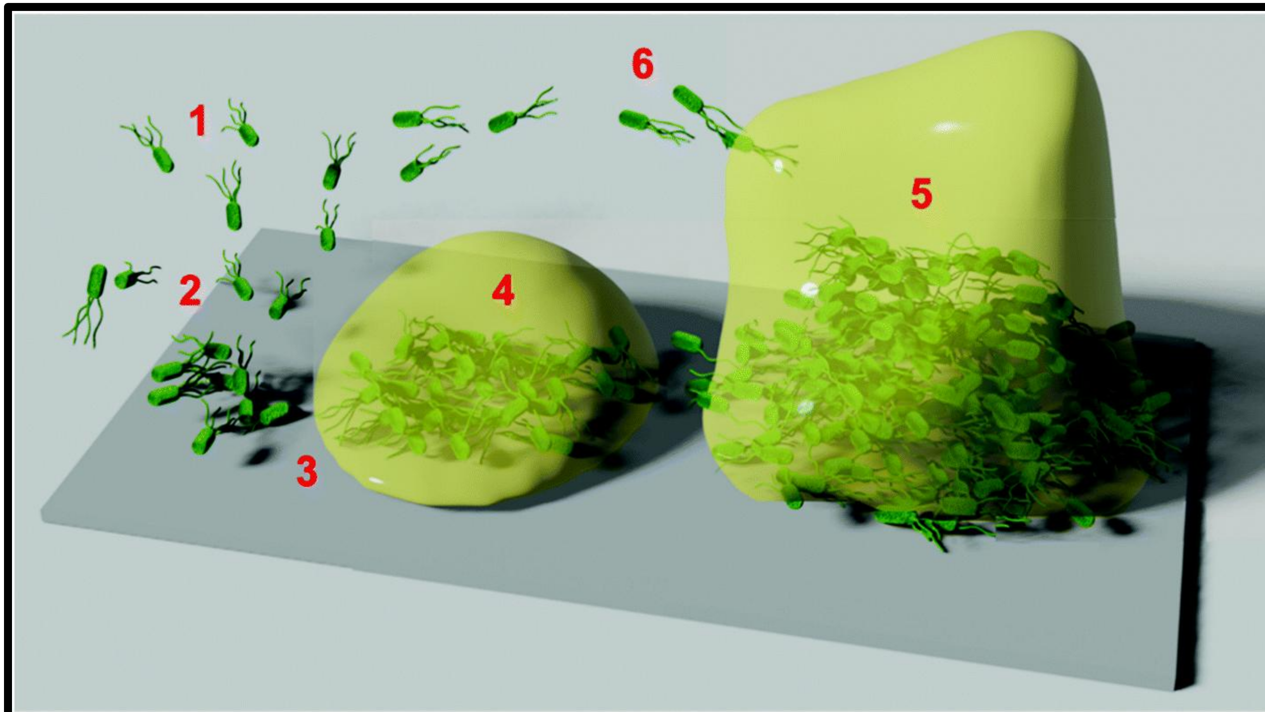
Examples of bacterial specific adherence to host cells or tissue

Bacterium	Bacterial adhesin	Attachment site
Streptococcus pyogenes	Cell-bound protein (M-protein)	Pharyngeal epithelium
Streptococcus mutans	Cell- bound protein (Glycosyl transferase)	Pellicle of tooth
Streptococcus salivarius	Lipoteichoic acid	Buccal epithelium of tongue
Bordetella pertussis	Fimbriae ("filamentous hemagglutinin")	Respiratory epithelium
Vibrio cholerae	N-methylphenylalanine pili	Intestinal epithelium
Mycoplasma	Membrane protein	Respiratory epithelium

3-Biofilm formation:

Some of the indigenous bacteria are able to construct biofilms on a tissue surface, or they are able to colonize a biofilm built by another bacterial species.

Biofilms usually occur when one bacterial species attaches specifically or non specifically to a surface, and then secretes carbohydrate slime that imbeds the bacteria and attracts other microbes to the biofilm for protection or nutritional advantages.



1. Planktonic bacteria
2. Initial reversible attachment followed by irreversible attachment
3. Formation of microcolonies and significant secretion of EPS to form biofilm matrix
4. Proliferation to form immature biofilm
5. Biofilm restructuring and maturation
6. Dispersal to colonize new regions

Importance of The Normal Flora (Advantages):-

1. They constitute a protective host defense for e.g (oral flora contribute to immunity by inducing low levels of circulating & secretory Ab that may cross react with pathogen.

2. They produce vitamin B and vitamin K in intestine.

3. The normal flora may antagonize other bacteria through the production of substances which inhibit or kill nonindogenous species

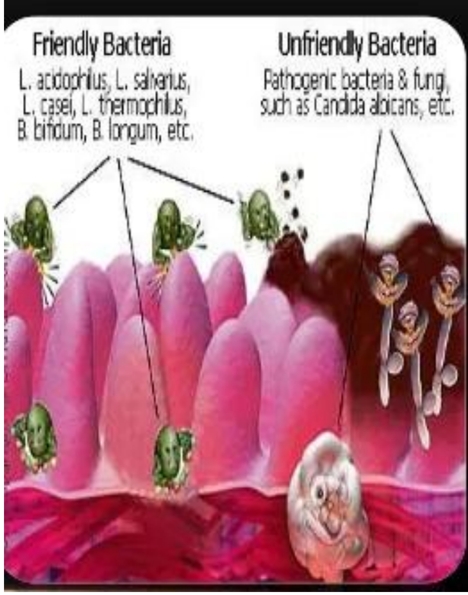
4. The oral bacteria flora exert microbial antagonism against nonindigenous species by production of inhibitory fatty acids, peroxides, bacteriocins, etc.

5. Recent research suggests the microbiota has an

inhibitory effect on the development of cancer of the colon

7. Normal flora aid in digestion of food

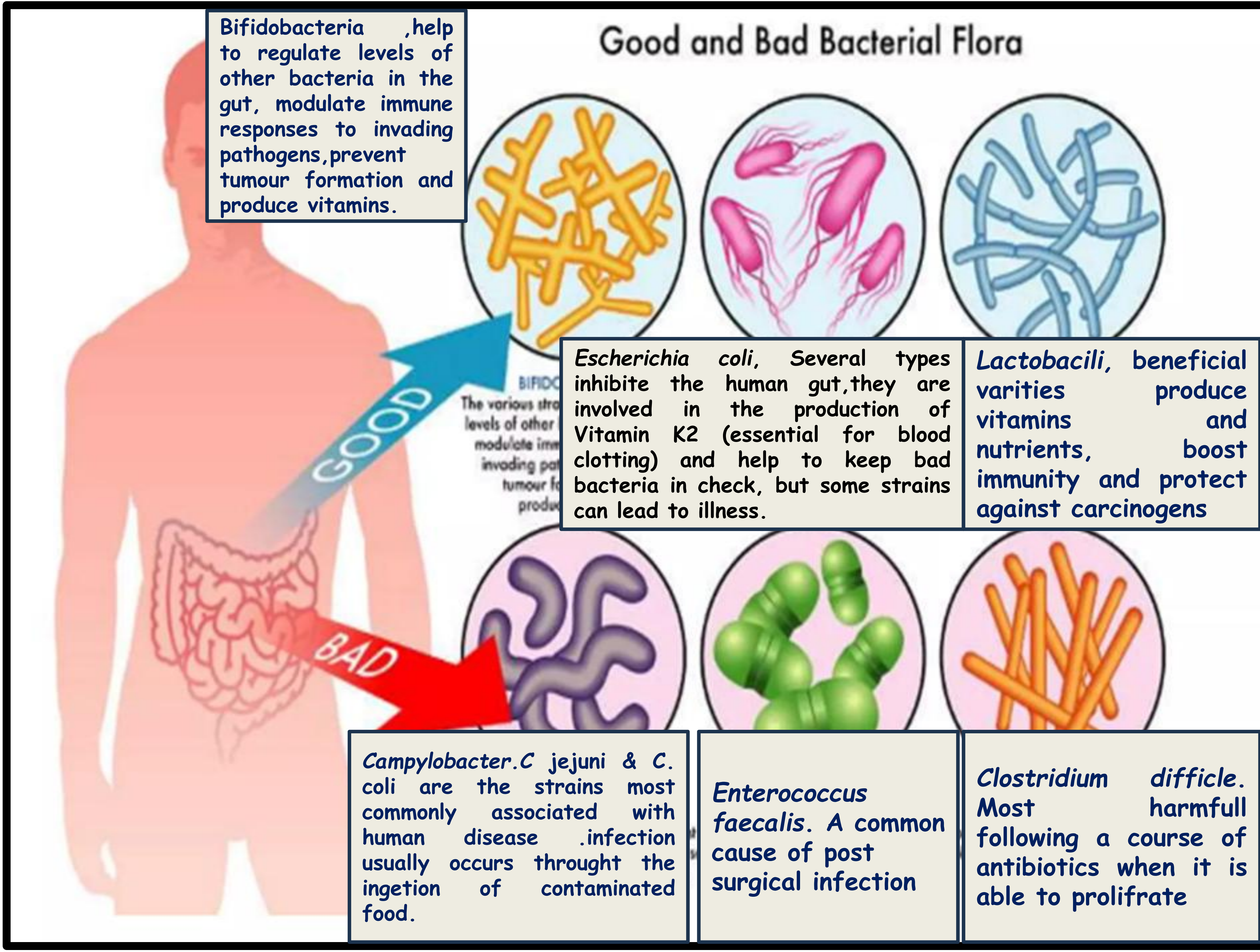
Disadvantages of the Normal Flora



1. They can cause disease in the following:
   a) When individuals become immunocompromised or debilitated.
   b) When they change their usual anatomic location.

2. The oral flora of humans may harm their host since some of these bacteria are pathogens or opportunistic pathogens

3. Production of carcinogens: Some normal flora may modify through their enzymes some chemicals in our diets into carcinogens e,g. Artificial sweeteners may be enzymatically modified into bladder carcinoges

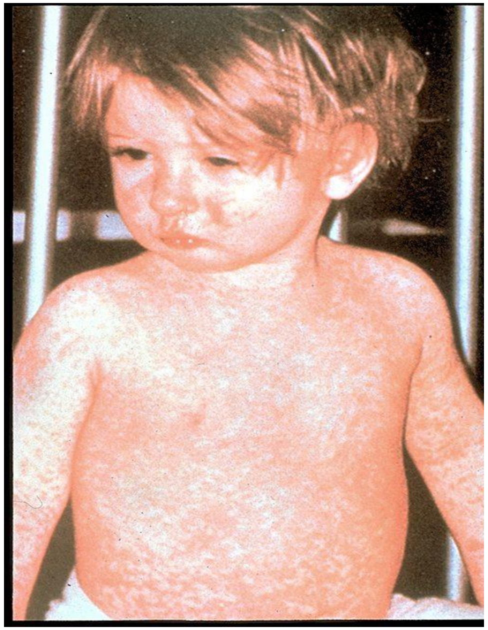

Good and Bad Bacterial Flora
Bifidobacteria ,help to regulate levels of other bacteria in the gut, modulate immune responses to invading pathogens,prevent tumour formation and produce vitamins.
GOOD
Escherichia coli, Several types inhibite the human gut,they are involved in the production of Vitamin K2 (essential for blood clotting) and help to keep bad bacteria in check, but some strains can lead to illness.
Lactobacili, beneficial varities produce vitamins and nutrients, boost immunity and protect against carcinogens
BAD
Campylobacter.C jejuni & C. coli are the strains most commonly associated with human disease .infection usually occurs throught the ingetion of contaminated food.
Enterococcus faecalis. A common cause of post surgical infection
Clostridium difficle. Most harmfull following a course of antibiotics when it is able to prolifrate

Normal Flora of the Skin

- The most important sites are:

1. Axilla
2. Groin
3. Areas between the toes

-The majority of skin microorganisms are found in the most superficial layers of the epidermis and the upper parts of the hair follicles.

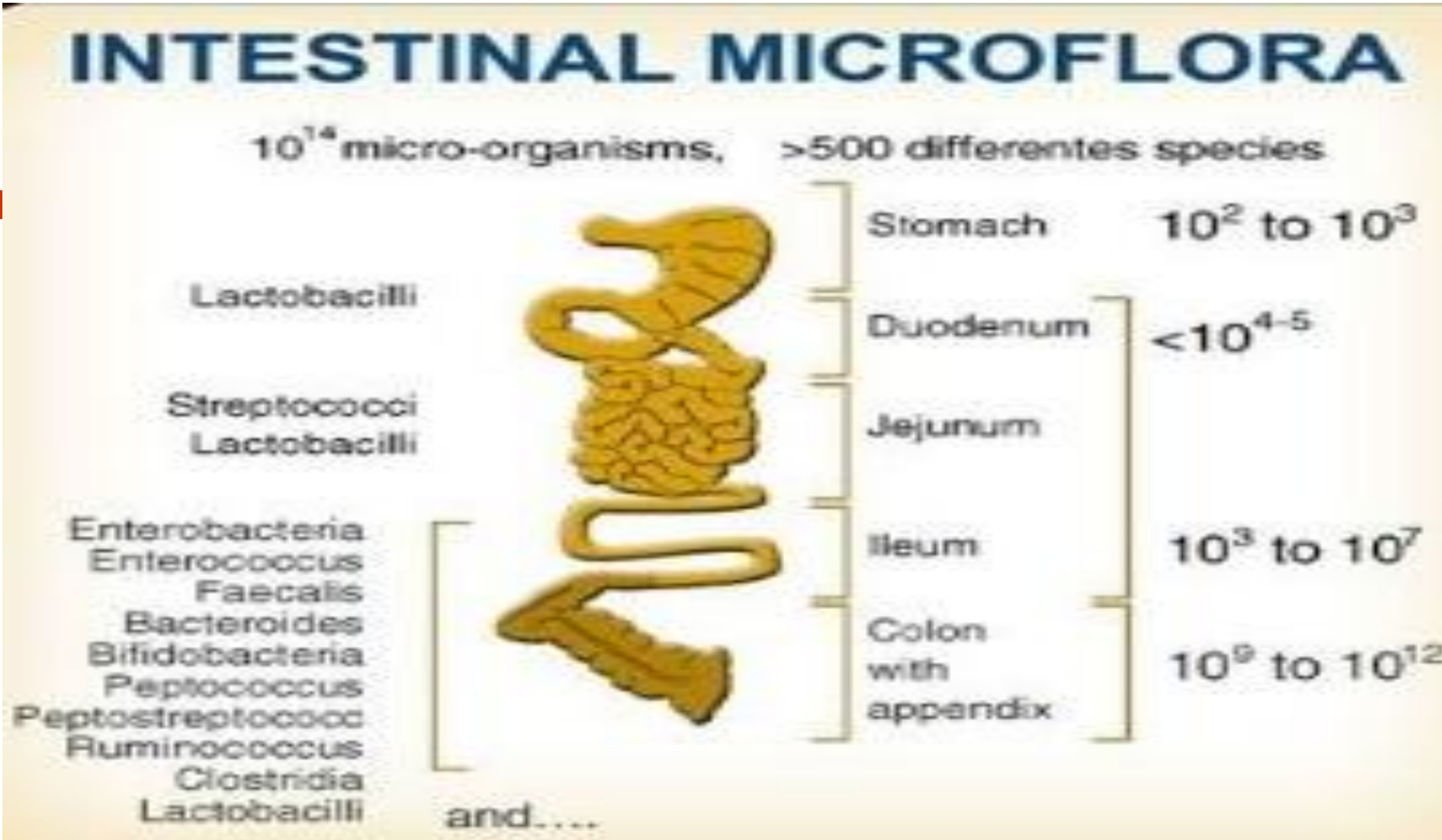


-Important bacteria & fungi of skin:

1. *Staphylococcus epidermidis*
2. *Micrococcus* sp.
3. *Corynebacteria* sp.
4. *Mycobacterium smegmatis*
5. *Candida albicans*
6. *Malassezia spp.*

Normal flora - Oral cavity

- Ecology and developmental stages
  - Birth: sterile mouth within 4-12 hours (lactobacilli, streptococci)
  - Neonate (Streptococcus salivarius, staphylococci, Neisseriae, Moraxella catarrhalis
  - Teeth appear (Streptococcus mutans, Streptococcus parasanguis)
  - Gingival crevice area (Anaerobic species, yeasts)
  - Puberty (Bacteroides, spirochetes)
- 10^8 bacteria/mL of saliva; potentially >700 species



- **Small intestine (10^3 -10^8)**
- **Large intestine (10^9-10^{12}/ml)**
    - **-Adult excretes $3x10^{13}$ bacteria/day(25%-35% of fecal mass = bacteria)**

Normal Flora of the GIT

- **Early gut colonization has four phases At birth :**

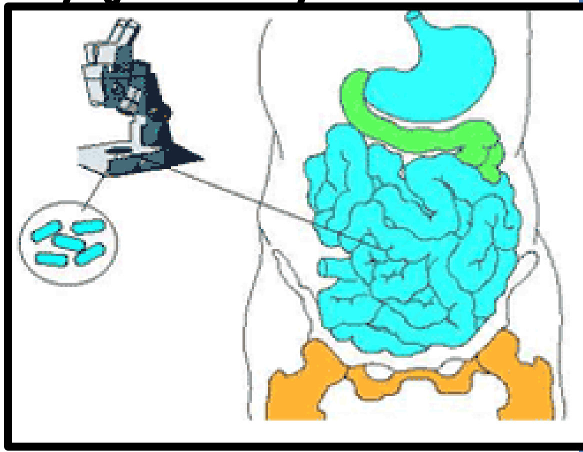
- Phase 1: The entire intestinal tract is sterile
- Phase 2: Initial acquisition ,vagina, feces, hospital.
- Phase3:, bacteria enter with the first feed, the initial bacterial
- colonization vary with the food source of the infant.

1-Breast feeding:, *Bifidobacteria* account for more than 90% of the total intestinal flora.

2-Bottle fed: more diverse more *Bacteroides* and *Clostridial* species.

Phase 4: Start of solids food move to adult flora (Fermicutes and bacterioteds), Bifidobacteria are progressively joined by other bacteria.

- *Enterobacteriaceae*
- *Enterococci* .
- *Bacteroides*
- *Staphylococci*
- *Lactobacilli*
- *Clostridia*



Gut microbiome origins

HOW DO WE GET OUR MICROBIOME?

BIRTH:

A newborn gets its microbes from:

- its mother's birth canal
- skin of its mother and other care-givers

BREAST MILK:

Breast milk has been fine-tuned over millions of years to provide:

- nutrients, vitamins, and antibodies
- diverse microbes to populate the baby's gut

ENVIRONMENT:

For the rest of the baby's life, it will continuously encounter new microbes from:

- soil and water
- people, pets, plants
- new and diverse foods

Normal Flora of the GIT

In the upper GIT of adult humans

- mainly acid-tolerant lactobacilli

e.g. Helicobacter pylori

- Around 10% of our body's energy is produced by our microbiota fermenting food in the gut.

Gut Microbiota in Health- symbiosis

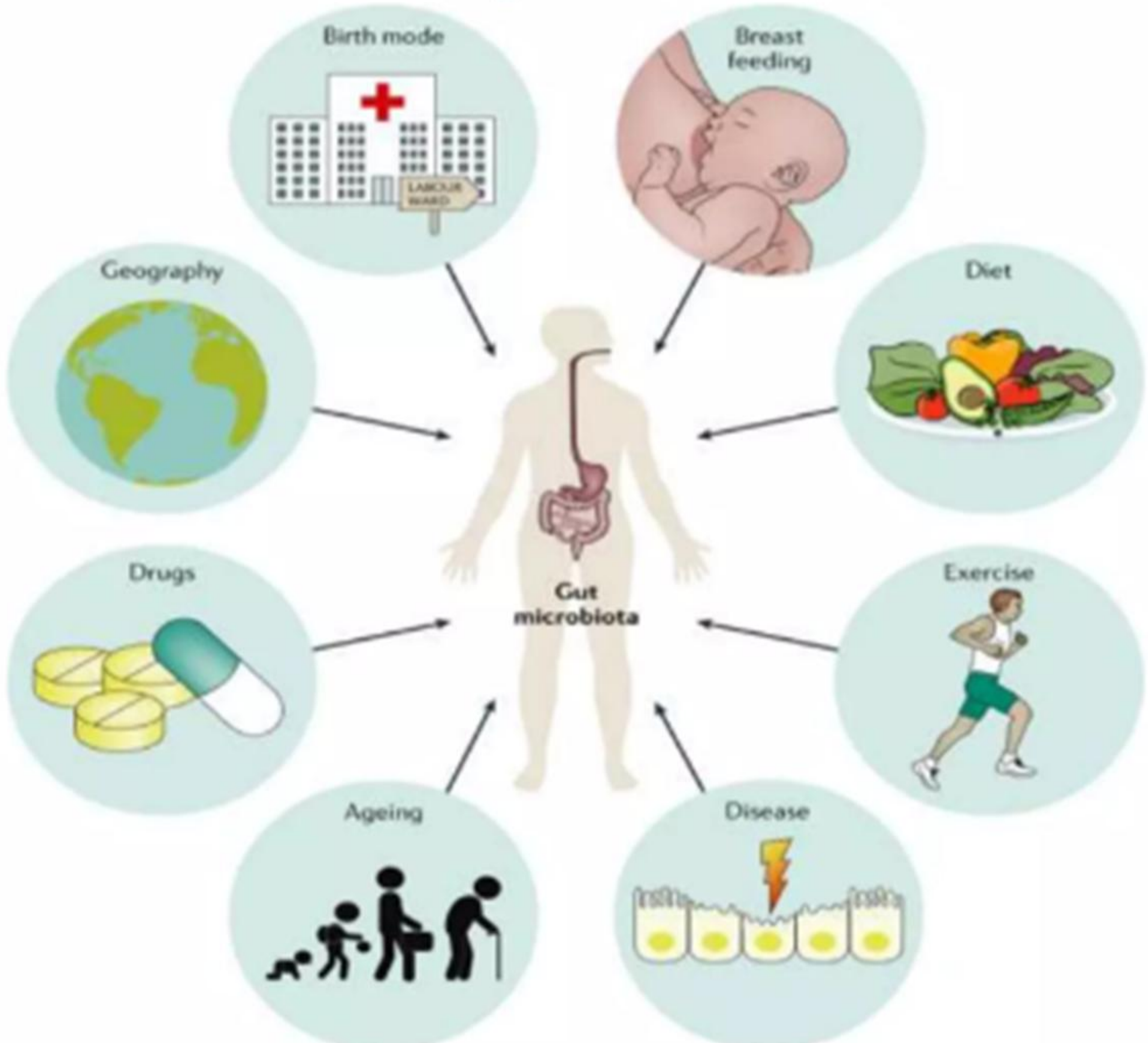
Provides:
- Nutrients
- Shelter

Human (host)

Microbiome

- Increases the metabolic capacity of the host -digest plant carbohydrates, milk products (glycans) - ~10% of the calories from diet
- Provide vitamins (e.g. B2, B12, K and folic acid)
- Protect from colonization with pathogenic bacteria (Colonization resistance)

Factors that can influence the composition and function of the human gut microbiota



Gut microbiota and modern lifestyle:

- Changes in human ecology have affected the composition of microbiota during human evolution but are more radical change has occured in recent decades.

- One of the most significant finding is that in developed countries there has been a loss of certain species that colonized our bowel some decades ago with the resultant loss of biodiversity of our microbiota.

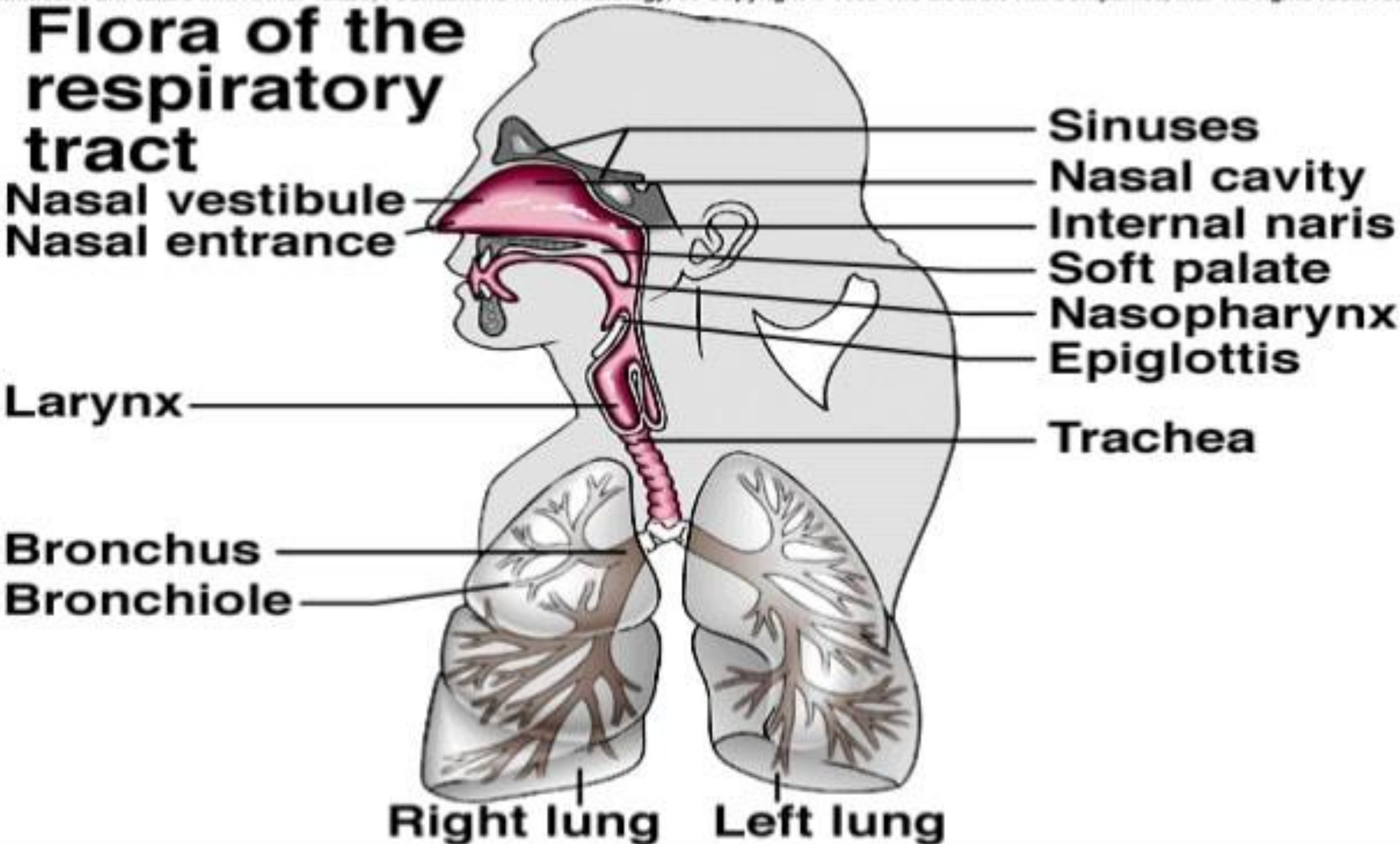
Factors that have influenced this change in microbiota include:-

-Water sanitation
-Increased performance of cesarean section
-More frequent use of antibiotics in preterm newborns,
-Decreased breast-feeding.
-Increased hygiene, or
-Widespread use of antibacterial soaps.

Normal flora - Conjunctiva

- Variety of bacteria: low numbers present
  - High moisture
  - Blinking mechanically removes bacteria
  - Lachrymal secretions include lysozyme

UPPER RESPIRATORY TRACT – AND LOWER RESPIRATORY TRACT



Normal flora of Respiratory tract

- Uper respiratory tract

- Nostrils:
  - *Staphylococcus spp corynebacterium*

- Nasopharynx :
  - α and β *Streptococcus*
  - *Neisseria spp.*
  - *Haemophilus influenzae*

- Oropharynx:
  - *Staph spp, coryne*
  - *α Strep, Neisseria spp*

Lower respiratory tract (trachea, bronchi, pulmonary tissues)

-Usually sterile
-Ciliated epithelium
-Mucus blanket: entrapment
-Alveolar macrophages

-If breached: opportunistic infections

- The person may become susceptible to infection by pathogens descending from the nasopharynx. e.g *H.influenzae & S. pneumoniae*

Normal flora of Urogenital tract

- Upper urinary tract (kidneys, ureters, bladder) usually sterile
- Male anterior urethra Same as skin +enteric+enterococcus

- Vagina: complex microbiota
  - At birth Same as mother
  - Neonate Same as skin+enteric+ strept
  - At puberty Lactobacillus+same as skin+anaerobes+strep
  - At menopause: return to prepuberty flora

  - Lactobacillus in vagina of adults has protective role against certain pathogen e.g
  - *Staph.epidermidis.*
  - *Neisseria sp.*
  - *Streptococcus agalactiae*

Normal Flora-human relationships

- **Mutualistic:-**

- *Escherichia coli* ,synthesizes **Vitamin K & B complex Vitamins.**
- In return, we provide a warm ,moist nutrient rich environment for ***E.coli***

- **Commensalistc:-**

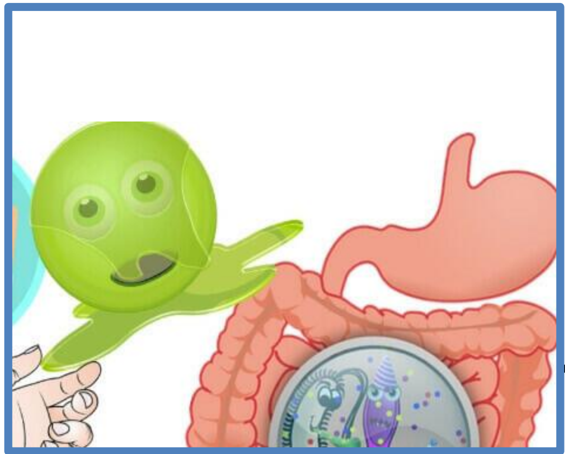
- We have no commensalistic relationships with bacteria.
- If bacteria are in or our body, they are either helping us (microbial antagonism) or harming us.

- **Opportunistic :-**

- *Escherichia coli –* normally in our digestive tract where it causes no problems, but if it gets into the urinary tract it can become pathogenic.
- *Staphylococcus aureus-* commonly found in upper respiratory tract ,but if it gets into a wound or a burn it can become pathogenic.
- -Host rendered susceptible by:
- -Immunosuppression, Radiation therapy & chemotherapy, Rheumatic heart disease

Types of Normal Flora

A/ Resident flora:



Microbes that are always present
1-acquired rapidly during & after birth.
2-Reflects nutrition of person.
3-Reflects genetics of person.
4-Reflects age of person.
5-Reflects environment of person.
6-Reflects sex of person

B/ Transient Flora:

Microbes that live in or on your body for a period of time (hours, days, weeks, months) then move on or die off.
-Cannot persist in the body ..because
1- Competition from other microorganisms
2- Elimination by the bodys defence cells.
3-Chemical or physical changes in the body

Thank you for your listening